

Hybrid Search & Retrieval POC

https://github.com/aiyshwary/Search_Engine

Python

FAISS

BM25

OpenSearch

Sentence Transformers

Tesseract

OVERVIEW

A production-style multilingual hybrid search engine for Indian-language documents, combining BM25 lexical search and FAISS semantic retrieval to rank documents with snippet-level evidence across Tamil, Hindi, Telugu and more.

IMPACT

Built a multilingual hybrid search pipeline supporting Indian-language PDFs and DOCX files, combining BM25 (OpenSearch) and FAISS IVF-PQ for accurate document retrieval with chunk-level evidence and best-snippet selection.

KEY DETAILS

- Ingests PDF and DOCX files with digital text extraction and Tesseract OCR fallback for scanned documents.
- Detects document language using FastText (LID176) across Tamil, Hindi, Telugu, Kannada, Bengali and more.
- Applies SimHash near-duplicate detection and token-aware chunking with sentence-based splitting.
- Generates multilingual embeddings using Sentence Transformers and indexes them in FAISS IVF-PQ.
- Hybrid BM25 and ANN retrieval with document-level score aggregation and best-snippet selection per result.